1) Calculator

Please write a program which asks the user for two numbers and an operation. If the operation is add, multiply or subtract, the program should calculate and print out the result of the operation with the given numbers. If the user types in anything else, the program should print out nothing.

Some examples of expected behaviour:

Number 1: 10
Number 2: 17
Operation: add

10 + 17 = 27

Number 1: 4
Number 2: 6
Operation: multiply

4 * 6 = 24

In [45]:
```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
operation = input("Enter operation (add, subtract, multiply): ")

if operation == "add":
    print(f"{num1}" " + " f"{num2}" " = " f"{num1+num2}")
elif operation == "subtract":
    print(f"{num1}" " - " f"{num2}" " = " f"{num1-num2}")
elif operation == "multiply":
    print(f"{num1}" " * " f"{num2}" " = " f"{num1*num2}")
```

```
4 * 6 = 24
```

In [ ]:

2) Daily wages

Please write a program which asks for the hourly wage, hours worked, and the day of the week. The program should then print out the daily wages, which equal hourly wage multiplied by hours worked, except on Sundays when the hourly wage is doubled.

Hourly wage: 8.5
Hours worked: 3
Day of the week: Monday
Daily wages: 25.5 euros

Hourly wage: 12.5
Hours worked: 10
Day of the week: Sunday
Daily wages: 250.0 euros

In [43]:
```
hourly_wages = float(input("Enter hourly wages: "))
hours_worked = float(input("Enter number of hours worked: "))
day_of_week = input("Enter day of the week: ")
day_of_week = day_of_week.lower()

if day_of_week == "sunday":
    daily_wages = hourly_wages * hours_worked * 2
else:
    daily_wages = hourly_wages* hours_worked

print(f"Daily wages = {daily_wages} euros" )
```

```
Daily wages = 25.5 euros
```

In [ ]:
```

```

3) Loyalty bonus

This program calculates the end of year bonus a customer receives on their loyalty card. The bonus is calculated with the following formula:

- If there are less than a hundred points on the card, the bonus is 10 %
- In any other case the bonus is 15 %
  The program should work like this:

How many points are on your card? 55
Your bonus is 10 %
You now have 60.5 points

But there is a problem with the program, so with some inputs it doesn't work quite right:

How many points are on your card? 95
Your bonus is 10 %
Your bonus is 15 %
You now have 120.175 points

Please fix the program so that there is always **either a 10 % or a 15 % bonus, but never both.**
Note: Keep the two if.

In [ ]:
```
# Fix the program
points = int(input("How many points are on your card? "))

if points < 100:
    points_new = points * 1.1
    print("Your bonus is 10 %")

if points >= 100:
    points_new = points * 1.15
    print("Your bonus is 15 %")
```

```
Your bonus is 10 %
You now have 104.50000000000001 points
```

4) FizzBuzz

Please write a program which asks the user for an integer number. If the number is divisible by three, the program should print out Fizz. If the number is divisible by five, the program should print out Buzz. If the number is divisible by both three and five, the program should print out FizzBuzz.

Some examples of expected behaviour:

Number: 9
Fizz

Number: 20
Buzz

Number: 7

Number: 45
FizzBuzz

In [ ]:
```
num = float(input("Enter a number: "))

if num % 3 == 0 and num % 5 == 0:
    print("FizzBuzz")
elif num % 5 == 0:
    print("Buzz")
elif num % 3 == 0:
    print("Fizz")
```

```
Buzz
```

OR another method:

In [39]:
```
number = float(input("Enter a number: "))

if number % 3 == 0:
    print("Fizz",end = "")

if number % 5 == 0:
    print("Buzz")
```

```
FizzBuzz
```

5) PIN and number of attempts

Please write a program which keeps asking the user for a PIN code until they type in the correct one, which is 4321. The program should then print out the number of times the user tried different codes.

PIN: 3245
Wrong
PIN: 1234
Wrong
PIN: 0000
Wrong
PIN: 4321
Correct! It took you 4 attempts

If the user gets it right on the first try, the program should print out something a bit different:

PIN: 4321
Correct! It only took you one single attempt!

In [ ]:
```
pin = ""
attempts = 1
while True:
```

```
    pin = int(input("Enter your PIN: "))
    if pin != 4321:
            print("Wrong!")
            attempts +=1


    elif pin == 4321:
        if attempts == 1:
            print("Correct! It only took you one single attempt!")
            break
        elif attempts > 1:
            print(f"Correct! It took you {attempts} attempts")
            break
```

```
Correct! It only took you one single attempt!
```

In [ ]:

```

```

6) Display numbers from -10 to -1

-10
-9
-8
-7
-6
-5
-4
-3
-2
-1

In [56]:

```
for i in range(-10,0,1):
    print(f"{i}")
```

```
-10
-9
-8
-7
-6
-5
-4
-3
-2
-1
```

7) Bonus: Reverse integer number

Given:

76592

Expected output:

29567

Note: Try to solve the problem with numbers, don't convert the numbers to strings. **Hint:** Use the modulo operator (%)

In [112…

```
num3 = int(input("Enter an integer number to reverse: "))

#859
revers = 0

while num3 != 0:
    temp = num3 % 10
    num3 //= 10
    revers =  revers* 10 + temp

print(f"{revers}")
```

```
29567
```